# Adaptive Resonance Theory 1 (ART1)

# Outline

1. Why ART?
2. Basic Idea of ART
3. ART1 Architecture
4. How ART1 Works
5. Vigilance Parameter
6. A Simple Example
7. Strengths and Limitations
8. Summary

# 1. Why ART?

Networks like the MLP are trained once on a fixed dataset. But in real life, **new data keep coming**. A learning system then needs two abilities:

**Stability**
Keep what has already been learned.
Old memories should not be destroyed.

vs

**Plasticity**
Stay flexible to learn new things.
New patterns should be learned quickly.

**The Stability–Plasticity Dilemma**
**Too plastic:** learning new data overwrites old knowledge ("catastrophic forgetting").
**Too stable:** the network cannot learn anything new.

# 1. Why ART?

**ART is inspired by how humans learn:** we can remember a new friend's face without forgetting our old friends.

**1976**

Stephen Grossberg proposes Adaptive Resonance Theory

**1987**

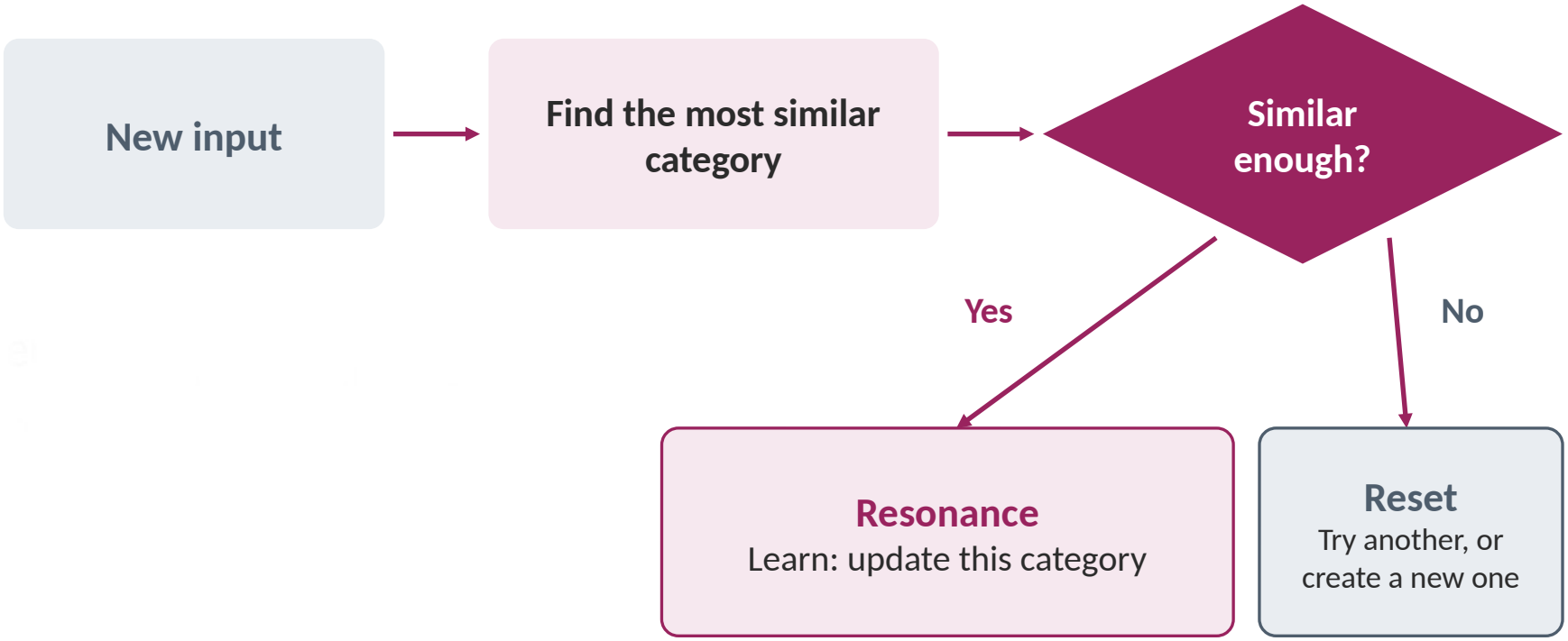
Carpenter & Grossberg build ART1, the first ART network (binary inputs)

**What does "resonance" mean?**

When the input and a stored memory are similar enough, the network "resonates": it accepts the match and learns. If they are not similar enough, it searches for another memory or creates a new one.

# 2. Basic Idea of ART

For every new input, ART asks one question: “Is this similar enough to something I already know?”




**Key point:** a new category is created only when the input is really new, so old categories are protected (stability) while new patterns can still be learned (plasticity).

# 2. Basic Idea of ART

## Main features of ART1

### 1 Binary inputs

ART1 works with binary patterns: every input is a vector of 0s and 1s.

### 2 Unsupervised & online

No labels needed. It groups similar inputs into clusters, learning one input at a time.

### 3 Flexible number of clusters

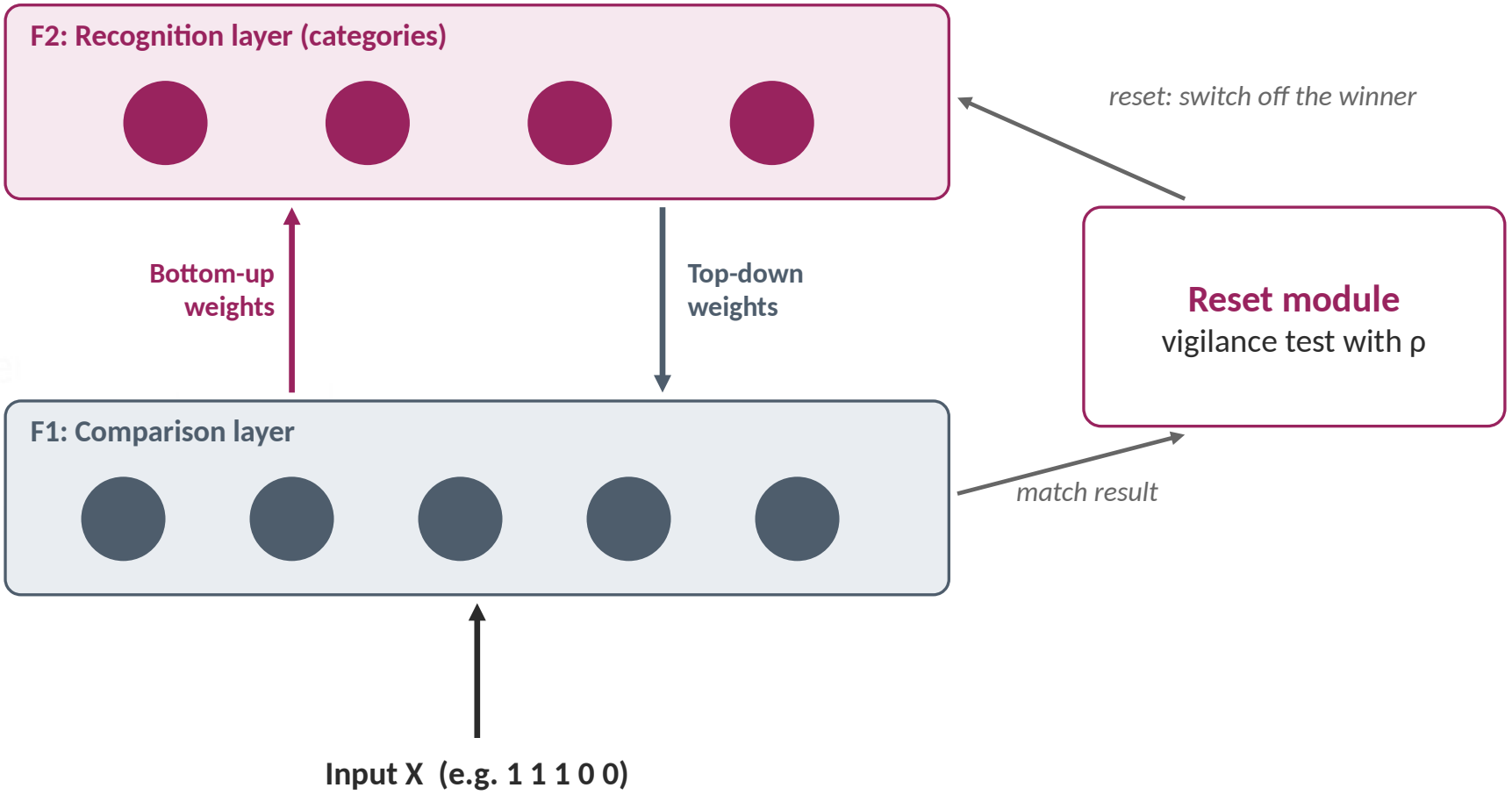
New categories are added when needed, unlike K-means where the number K is fixed in advance.

### 4 One key parameter

The vigilance ρ controls how strict the similarity check is.

# 3. ART1 Architecture

ART1 is the simplest ART network. It works with binary inputs (0/1).



# 3. ART1 Architecture

## What does each part do?

| | |
|---|---|
| **F1 layer** | Comparison layer. Receives the input X and compares it with the stored pattern of the winning category. |
| **F2 layer** | Recognition layer. Each neuron = one category. Neurons compete and the winner takes all. |
| **Weights** | Bottom-up weights help choose the winner. Top-down weights store the category's pattern (called the template). |
| **Reset** | Checks the match using vigilance ρ. If the match is too low, it switches off the winner and the search continues. |

# 4. How ART1 Works

| | | |
|---|---|---|
| 1 | **Input** | Present a binary input X to layer F1. |
| 2 | **Recognition** | F2 neurons compete. The most similar category J wins. |
| 3 | **Comparison** | Compare X with the template T (stored pattern) of the winner J. |
| 4 | **Vigilance test** | Is the match ≥ ρ ? |
| 5 | **Yes → Resonance**<br>Update T to keep only the features shared by X and T. Done. | **No → Reset**<br>Switch off J and go back to step 2. If no category passes, create a new one. |

# 4. How ART1 Works

**How do we measure “similar enough”?**

$$\text{Match} = |X \text{ AND } T| \;/\; |X|$$

(number of 1s shared by X and T) / (number of 1s in X)

**X** = the input pattern

**T** = the template of the winning category: the 0/1 pattern this category remembers

| | | | | | |
|---|---|---|---|---|---|
| **Input X** | 1 | 1 | 1 | 0 | 0 |
| **Template T** | 1 | 1 | 0 | 0 | 0 |
| **X AND T** | 1 | 1 | 0 | 0 | 0 |

**Match = 2 / 3 = 0.67**

If ρ = 0.5 → pass (resonance)

If ρ = 0.8 → fail (reset)

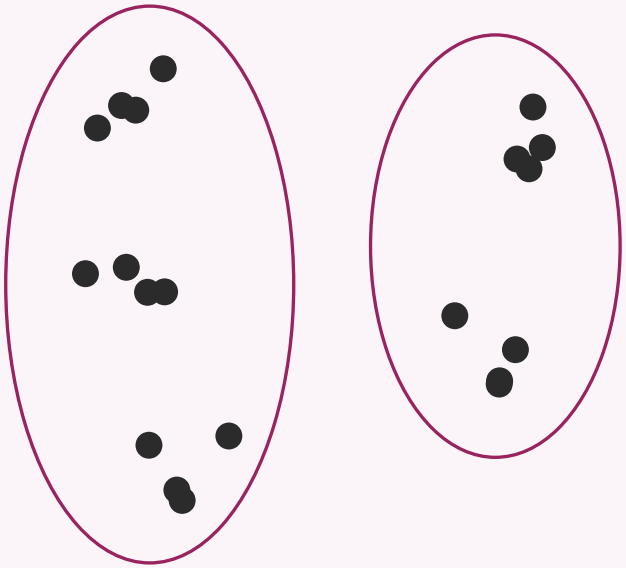
**Learning after resonance:** **new T = X AND T**

The template keeps only the 1s that both patterns share, so each category remembers the common features of its members.

# 5. Vigilance Parameter ρ

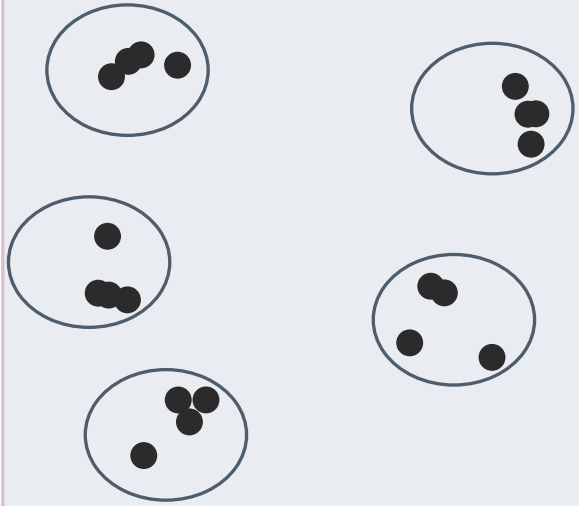
ρ is a number between 0 and 1 chosen by the user. It decides how **picky** the network is.

**Low ρ (e.g. 0.3)**



Loose check → few, large categories
More general

**High ρ (e.g. 0.9)**



Strict check → many, small categories
More detailed

# 6. A Simple Example

Four 5-bit input patterns are presented one by one:




**Rules (from the previous slides)**

Match = |X AND T| / |X|

Match ≥ ρ → resonance, and new T = X AND T

No category passes → create a new category with T = X

**Run 1: ρ = 0.5**

**Run 2: ρ = 0.8**

# 6. A Simple Example (ρ = 0.5)

| Input | Compare with | Match | Decision | Categories after |
|---|---|---|---|---|
| P1 = 11100 | (no category yet) | – | Create C1 | C1 = 11100 |
| P2 = 11000 | C1 = 11100 | 2/2 = 1.00 ✓ | Resonance, update C1 | C1 = 11000 |
| P3 = 00011 | C1 = 11000 | 0/2 = 0.00 × | Reset, create C2 | C1 = 11000<br>C2 = 00011 |
| P4 = 00111 | C2 = 00011 | 2/3 = 0.67 ✓ | Resonance, update C2 | C1 = 11000<br>C2 = 00011 |

**Result: 2 categories {P1, P2} and {P3, P4}**

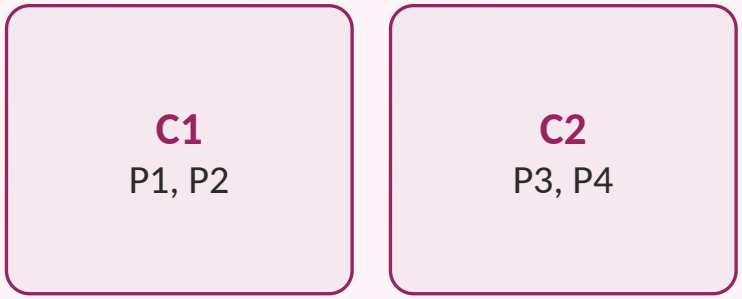
Note: after P2, C1 becomes 11000 — it keeps only the common 1s of P1 and P2.

# 6. A Simple Example (ρ = 0.8)

P1, P2 and P3 give the same result as before. The difference is P4:

| Input | Compare with | Match | Decision |
|---|---|---|---|
| P4 = 00111 | C2 = 00011 | 2/3 = 0.67 < 0.8 × | Reset, try next |
| | C1 = 11000 | 0/3 = 0.00 < 0.8 × | Reset, create C3 = 00111 |

**ρ = 0.5 → 2 categories**




**ρ = 0.8 → 3 categories**



# 7. Strengths and Limitations of ART1

## Strengths

- Learns new patterns without forgetting old ones
- Learns online, one input at a time
- Number of categories does not need to be set in advance
- Simple and fast to compute

## Limitations

- Only works with binary (0/1) inputs
- The result depends on the order of the inputs
- Choosing a good ρ is not always easy
- Templates can only lose 1s (AND), so they may become too small

Later models such as ART2 and Fuzzy ART extend these ideas to real-valued inputs.

# 8. Summary

| | K-means | ART1 |
|---|---|---|
| Input | Real-valued vectors | Binary vectors (0/1) |
| Learning | Unsupervised clustering | Unsupervised clustering + vigilance test |
| Number of clusters | K fixed in advance | Grows when needed |
| New data | Re-run on all data | Learns online, one at a time |
| Key parameter | K | Vigilance ρ |

## Take-home messages

ART1 solves the stability–plasticity dilemma for binary patterns.
Match ≥ ρ → resonance and learn; otherwise reset and search, or create a new category.
Higher ρ → more and finer categories; lower ρ → fewer and broader categories.

# Thank You !

# 9. Reference

[1] S. Grossberg, “Adaptive pattern classification and universal recoding: I. Parallel development and coding of neural feature detectors,” Biological Cybernetics, vol. 23, pp. 121–134, 1976.

[2] G. A. Carpenter and S. Grossberg, “A massively parallel architecture for a self-organizing neural pattern recognition machine,” Computer Vision, Graphics, and Image Processing, vol. 37, pp. 54–115, 1987.

[3] G. A. Carpenter and S. Grossberg, “The ART of adaptive pattern recognition by a self-organizing neural network,” Computer, vol. 21, no. 3, pp. 77–88, 1988.

[4] L. Fausett, Fundamentals of Neural Networks: Architectures, Algorithms, and Applications, Prentice Hall, 1994 (Chapter 5).

[5] https://en.wikipedia.org/wiki/Adaptive_resonance_theory